Tutorial 4 (to be completed in week4)
Normalisation

Tutorial Objectives:

- Understand normalisation and its various forms (1NF, 2NF, 3NF)
- Use Dependency Diagrams to arrive at a good set of tables (relational schema). Apply concepts of partial and transitive dependencies on a given data set.
- Draw a relational diagram (e.g. ERD)
- Create and populate tables in Oracle based on the above diagram (ERD)

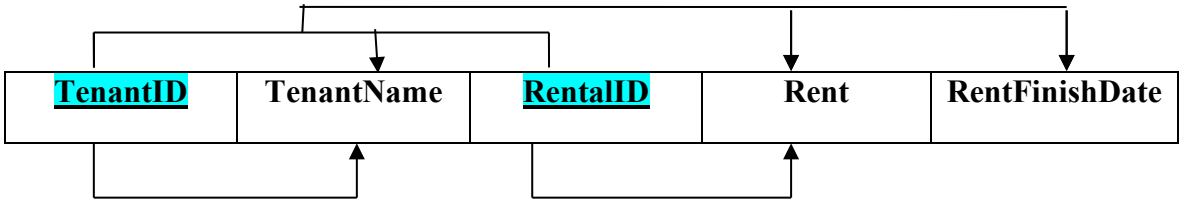
Tutorial Tasks:

Answer the following questions about normalisation:

1. What is normalisation?

2. Is it always necessary to normalise data? Can you run an application with un-normalised data?

3. Given the dependency diagram shown below, answer 3a-3c.



   a. Identify and discuss each of the indicated dependencies.

   b. Create a database whose tables are at least in 2NF, showing the dependency diagrams for each table.

   c. Create a database whose tables are at least in 3NF, showing the dependency diagrams for each table.

4. Using the INVOICE table structure shown in Table P6.3*, write the relational schema, draw its dependency diagram and identify all dependencies, including all partial and transitive dependencies.

   Assumptions:
   a. The table does not contain repeating groups.
   b. Any invoice number may reference more than one product (i.e. each invoice usually has more than one product, e.g. invoice number 211347 is referring to three products AA-E3422QW, QD-300932X, RU-995748G). (*Hint*: This table uses a composite primary key.)

c. Any given product is supplied by a single vendor, but a vendor can supply many products (e.g. product AA-E3422QW is supplied only by the vendor 211 but vendor 211 is supplier for two products (also supplies QD-300932X).

Table P6.3 Sample INVOICE Records

Attribute Name	Sample Value	Sample Value	Sample Value	Sample Value	Sample Value
INV_NUM	211347	211347	211347	211348	211349
PROD_NUM	AA-E3422QW	QD-300932X	RU-995748G	AA-E3422QW	GH-778345P
SALE_DATE	15-Jan-2018	15-Jan-2018	15-Jan-2018	15-Jan-2018	16-Jan-2018
PROD_LABEL	Rotary sander	0.25-in. drill bit	Band saw	Rotary sander	Power drill
VEND_CODE	211	211	309	211	157
VEND_NAME	NeverFail, Inc.	NeverFail, Inc.	BeGood, Inc.	NeverFail, Inc.	ToughGo, Inc.
QUANT_SOLD	1	8	1	2	1
PROD_PRICE	$49.95	$3.45	$39.99	$49.95	$87.75

5. Using the initial dependency diagram drawn in Problem 4, remove all partial dependencies, write the relational schema, and draw the new dependency diagrams, and identify the normal forms for each table structure you created.
   (*Hint:* PROD_NUM → PROD_LABEL, PROD_PRICE, VEND_CODE, VEND_NAME)
   (*Hint*: Your actions should produce three dependency diagrams.)

6. Using the table structures created in Problem 5, remove all transitive dependencies, and draw the new dependency diagrams. Also, identify the normal forms for each table structure you created.

7. Using the results of Problem 6, draw the Crow's Foot ERD (or an equivalent diagram).

8. Identify appropriate Oracle datatype for each attribute given in the Invoice table (Refer to Week4 Lecture slides 54-55 for Datatypes).

9. Implement the ERD by writing and executing the Oracle scripts in your Oracle schema.

10. Insert at least two records into each of the tables.